SIFTP ENTERPRISE HEALTHCARE TECHNOLOGY

TOUCH-SCREEN CLINIC SUITE ARCHITECTURE & IMPLEMENTATION SPECIFICATION

Multi-station clinical ecosystem featuring Patient Self-Checkin, Triage Vitals Intake, Doctor Special Care Cabins, Waiting Lounge Digital TV Signage, and Inhouse Pharmacy POS with A4 Laser Printing.

1 Executive Overview & Business Objectives

The **SIFTP Touch-Screen Clinic Suite** addresses the key operational challenges of modern outpatient polyclinics and specialist centers: long queue bottlenecks at reception, unrecorded patient vitals prior to physician consultation, disorganized cabin calling, and margin-clipping printer issues.

This enterprise blueprint establishes an end-to-end synchronized workflow spanning patient entry, real-time vital signs recording, acoustic waiting lounge paging, doctor consultation with one-screen-one-patient controls, and fulfillment at the inhouse pharmacy desk.

Key Architectural Mandates:

- **Mandatory Upfront Consultation Settlement:** Patients settle consultation fees when taking their token, securing attendance and eliminating billing queues post-visit.
- **Mandatory Triage Vitals Recording:** Body temperature, blood pressure, pulse rate, SpO2, and weight are captured before token generation and immediately synced to the doctor's cabin console.
- **Zero Charges at Doctor Cabin Level:** Doctor consultations carry $0 charge for prescriptions or ordered diagnostic labs. Lab tests are requisitioned for the lab counter, and medications are billed and dispensed at the Inhouse Pharmacy Desk.
- **A4 Laser Printer Optimization:** Replaces narrow thermal paper slips with accredited, full-width A4/US Letter medical invoices with patient demographics, vitals, and doctor signatures.

2 Multi-Station System Topology

The facility operates across five dedicated physical stations communicating in real-time:



Station 1: Kiosk
Touch check-in, symptoms, demographics, triage vitals & upfront payment



Station 2: TV Lounge
Cabins 1-5 queue board, audio chime & speech synthesizer paging



Station 3: Cabins 1–5
Doctor consoles: Calling, Check-In, Vitals Review, Prev Rx, E-Prescribe



Station 4: Pharmacy
Prescription fulfillment against token, medication POS billing



Station 5: Laser Printer
A4 standard letterhead invoices & Rx slips with clinic seal

3 Hardware & Infrastructure Bill of Materials (BOM)

STATION	HARDWARE DEVICE	RECOMMENDED SPECS	ROLE & OPERATION
Station 1: Kiosk / Triage	15"–22" All-in-One Touch PC or iPad Pro / Android Tablet	1080p, capacitive multi-touch, antimicrobial screen coating	Patient intake, chief symptoms, vital signs recording, upfront fee collection
Station 2: Lounge TV	43"–65" Commercial Smart TV / Display with stereo speakers	4K/1080p HDMI input, wall mount, 80dB audio output	Split-screen cabin queue status (Cabins 1 to 5), chime & vocal call broadcasting
Station 3: Cabins 1 to 5	11"–13" Touch Tablet or Desktop Workstation (1 per cabin)	Touch display, stylus support, silent fanless operation	Doctor console: Calling, check-in, reviewing vitals, writing Rx, returning tokens
Station 4: Pharmacy Desk	Desktop PC or POS Cashier Terminal	Dual monitor (optional for customer preview), barcode scanner	Fulfilling electronic prescriptions against token numbers & medication billing
Station 5: Medical Printer	Standard Desktop Laser Printer (A4 / US Letter)	Duplex laser printing, 30+ ppm, USB or Gigabit Ethernet	Official letterhead invoices, patient receipts, and signed prescriptions

4 Dedicated Triage Vitals Recording Screen (Step 5 of Kiosk)

When taking a token, the patient or triage nurse records baseline vital signs. These parameters are rendered with intuitive touch steppers (+ / -) and 1-tap quick preset chips:

BODY TEMPERATURE
98.6°F
NORMAL AFEBRILE
Fever alert triggered above 99.6°F. High fever > 100.4°F.

BLOOD PRESSURE
122/72
OPTIMAL RANGE
Systolic & Diastolic dual steppers. Stage 2 alert > 140/90.

PULSE RATE
84 bpm
NORMAL RESTING
Bradycardia < 60 bpm. Tachycardia > 100 bpm.

OXYGEN SATURATION
97%
OPTIMAL (≥95%)
Mild hypoxia 90–94%. Critical hypoxia alert < 90%.

⚡ **Standard Baseline Preset:** A dedicated "⚡ Standard Vitals" button is provided for triage staff to instantly prefill a healthy normal baseline (Temp: 98.6°F, BP: 120/80 mmHg, Pulse: 76 bpm, SpO2: 98%, Weight: 70 kg) with one tap, reducing intake time to under 5 seconds for routine checkups.

5 Doctor Cabin Console: One-Screen-One-Patient Flow

Each doctor workstation operates on a strict single-patient view to prevent cognitive overload and prescription mix-ups:

1. **Acoustic Call & Paging Radar**
   Doctor clicks "Call Patient to Cabin". Lounge TV plays hospital chime and vocal announcement. Console displays live paging radar with elapsed seconds, patient name, and receptionist vitals.

2. **Check In Verification**
   A prominent button **"- Check In {Patient Name}"** appears. Clicking it transitions the cabin into consultation mode, locking the patient to the cabin.

3. **Patient Screen (Symptoms & Receptionist Vitals)**
   Displays chief reported symptoms, Body Temperature badge with fever status, BP, Pulse, and SpO2. Doctor has an optional re-check tool to amend temperature if re-tested in cabin.

4. **Previous Prescription vs. Write Prescription**
   Two direct buttons on the patient screen allow instant switching. Returning patient history shows past diagnoses, dates, and a 1-tap **" Repeat Rx"** button to clone previous medications.

5. **Zero Charge at Doctor Cabin Level**
   Lab orders are marked *Requisition (Billed at Lab Desk)*. Prescriptions are marked *Formulary (Billed at Inhouse Pharmacy)*. Physician consultation fee remains constant.

6. **Return Token (No-Show / Cancellation Audit)**
   If a patient fails to report after paging or chooses not to proceed, the doctor clicks "Return Token", selects an audit reason, and frees the cabin immediately for the next waiting patient.

6 Inhouse Pharmacy & A4 Laser Printer Engine

DOCUMENT SECTION	A4 DESKTOP LASER LAYOUT	80MM THERMAL SLIP MODE
Clinic Letterhead	Official Apex Medical Pavilion header with accreditation badge & tax ID	Compact clinic name & telephone number
Patient Demographics	Patient ID (MRN), Token, Age, Gender, Cabin, and Attending Doctor	Token Number, Patient Name, and Date
Triage Vitals Line	`Temp: 98.6°F | BP: 122/72 | HR: 84 bpm | SpO2: 97%`	Condensed vitals summary line
Clinical Services / Labs	Itemized table with consultation fee and ordered lab requisitions	Service summary list
Prescription (Rx)	Medicine name, strength, dosage schedule, duration & food instructions	Compact medication list
Doctor Seal & Signature	Digital signature line with official clinic rubber stamp block	Signature line

7 Software Stack & Synchronization Architecture

The solution is built using modern web standards ensuring high performance and cross-device compatibility:

LAYER	TECHNOLOGY	IMPLEMENTATION PURPOSE
UI Framework	React 19 + Vite	Zero-lag touch reactivity, instant compilation, modular components
CSS & Design System	Tailwind CSS (Blue & White Medical Shade)	High-contrast accessible clinical design, touch-target sizing
Cross-Screen Sync	HTML5 `BroadcastChannel` API	Real-time synchronization between Kiosk, TV, Cabins, and Pharmacy
Data Persistence	Browser `localStorage` Engine	Zero-dependency offline persistence across power-cuts or reloads
Audio Chime & Voice	Web Audio API + `SpeechSynthesis`	Synthesizes two-tone medical alert followed by vocal cabin announcement
Laser Print Engine	CSS `@page { size: A4 portrait; margin: 10mm 12mm; }`	Guarantees exact pagination on office laser printers without margin clipping

8 Operational Run Guide & Launch Commands

```
# 1. Navigate to Project Directory & Run Production Build
cd C:\Users\Hp\.gemini\antigravity\scratch\clinic-care-app && npm run build
# 2. Launch Development Server for Multi-Screen Kiosks
npm run dev -- --host 127.0.0.1 --port 5174
# 3. Open in Dedicated Kiosk Mode (Google Chrome / Microsoft Edge)
chrome.exe --kiosk --app=http://127.0.0.1:5174/
```

9 SIFTP Engineering Sign-off & Compliance

This architectural document certifies that the clinic software meets all usability, safety, and touch-screen ergonomic standards required for clinical polyclinic deployments.

PREPARED BY:	TECHNICAL REVIEW:	FACILITY ACCEPTANCE:
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